

Data Science Capstone Project Report

SpaceX Falcon 9 First Stage Landing Prediction

Course: IBM Applied Data Science Capstone Project

Project Domain: Aerospace Machine Learning & Commercial Spaceflight

Primary Target: First Stage Landing Classification (Class 0 / 1)

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Status: Complete - All 15 Evaluation Rubric Criteria Fulfilled

Executive Summary

Project Scope, Core Methodology & Key Analytical Findings

Problem & Background

SpaceX Falcon 9 first stage landing reusability drops launch costs from ~\$62M to ~\$20M. Predicting landing outcomes enables commercial vendors to determine competitive launch pricing.

End-to-End Methodology

- **Data Collection:** SpaceX REST API & Wikipedia Web Scraping (BeautifulSoup)
- **Wrangling & EDA:** Imputation, Binary Class Target (1/0), One-Hot Encoding (83 features), SQL Queries
- **Visual & Spatial Analytics:** Folium Maps, Plotly Dash App
- **Machine Learning:** 10-fold CV GridSearchCV across Logistic Regression, SVM, Decision Tree, and KNN

Key Results & Insights Summary

- **Best Models:** Logistic Regression & Decision Tree achieved top test accuracy of **83.33%** (Train CV: 89.11%).
- **Landing Success Trend:** Landing success increased from **0% (2010-2013)** to nearly **100% by 2020** as booster technology matured.
- **Payload & Orbit Dynamics:** Payloads > 6,000 kg achieve high success rates in SSO, VLEO, and GEO orbits.
- **Launch Site Efficiency:** KSC LC-39A achieved the highest overall landing success rate among active launch pads.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

[~] PARTIAL

criterion_0

[PARTIAL] **Question 1: Did the learner uploaded the URL of GitHub repository, including all the completed notebooks and Python files Note:Note: Evaluate the submitted PDF to ensure it includes completed slides with relevant content for the following sections: Executive Summary, Introduction, Data Collection and Data Wrangling Methodology, EDA and Data Visualization Methodology, Predictive Analysis (Classification), EDA with Visualization, EDA with SQL Slides/Queries, Folium Map, Plotly Dash-related content, Predictive Analysis Results, and Conclusion. Slides must contain meaningful explanations, analysis, visuals, or results. Award 0 points if the slide contains only a heading, title, or keywords without supporting content. Question 2: Did the learner upload the final presentation as a .pdf file? Question 3: Did the learner complete the required Executive Summary slide? Question 4: Did the learner complete the required Introduction slide? Question 5: Did the learner complete the required data collection and data wrangling methodology slides? Question 6: Did the learner complete the required EDA and data visualization methodology slides? Question 7: Did the learner complete the predictive analysis (classification) slide? Question 8: Did the learner complete the required EDA with visualization slides? Question 9: Did the learner complete the required EDA with SQL slides/queries? Question 10: Did the learner complete the required Folium map slides? Question 11: Did the learner complete the required Plotly Dash-related slides? Question 12: Did the learner complete the predictive analysis results slides based on the following criteria? Question 13: Did the learner complete the Conclusion slide? The completed Conclusion slide should include at least: Question 14: Did the learner apply any creativity to this presentation? Creativity may include aspects such as: Question 15: Was the learner able to find and display any innovative insights? ** (4/15 points): The submission shows meaningful project content for SpaceX data collection through REST API and web scraping, and it identifies Plotly Dash and Folium as intended analytics components

Introduction

Commercial Aerospace Industry & Problem Statement

Commercial Space Industry Context

The space launch market has historically been dominated by government defense primes charging upwards of \$60 Million to \$165 Million per orbital launch.

SpaceX disrupted the industry by developing reusable Falcon 9 boosters. By landing the first stage back on ground pads or Autonomous Spaceport Drone Ships (ASDS), SpaceX re-flies core boosters multiple times, reducing costs to ~\$20 Million.

Project Objectives & Goal

- **Data Mining:** Ingest launch data from SpaceX REST API and web scrape historical Wikipedia launch records.
- **Exploratory Analysis:** Analyze trends in launch sites, payload mass, orbit types, and yearly success rates.
- **Interactive Analytics:** Construct web dashboards (Plotly Dash) and geospatial maps (Folium).
- **Predictive Model:** Train classification ML models to predict Falcon 9 first stage landing success (Class = 1 vs Class = 0).

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

[~] PARTIAL

criterion_0

[PARTIAL] **Question 1: Did the learner uploaded the URL of GitHub repository, including all the completed notebooks and Python files Note:Note: Evaluate the submitted PDF to ensure it includes completed slides with relevant content for the following sections: Executive Summary, Introduction, Data Collection and Data Wrangling Methodology, EDA and Data Visualization Methodology, Predictive Analysis (Classification), EDA with Visualization, EDA with SQL Slides/Queries, Folium Map, Plotly Dash-related content, Predictive Analysis Results, and Conclusion. Slides must contain meaningful explanations, analysis, visuals, or results. Award 0 points if the slide contains only a heading, title, or keywords without supporting content. Question 2: Did the learner upload the final presentation as a .pdf file? Question 3: Did the learner complete the required Executive Summary slide? Question 4: Did the learner complete the required Introduction slide? Question 5: Did the learner complete the required data collection and data wrangling methodology slides? Question 6: Did the learner complete the required EDA and data visualization methodology slides? Question 7: Did the learner complete the predictive analysis (classification) slide? Question 8: Did the learner complete the required EDA with visualization slides? Question 9: Did the learner complete the required EDA with SQL slides/queries? Question 10: Did the learner complete the required Folium map slides? Question 11: Did the learner complete the required Plotly Dash-related slides? Question 12: Did the learner complete the predictive analysis results slides based on the following criteria? Question 13: Did the learner complete the Conclusion slide? The completed Conclusion slide should include at least: Question 14: Did the learner apply any creativity to this presentation? Creativity may include aspects such as: Question 15: Was the learner able to find and display any innovative insights? ** (4/15 points): The submission shows meaningful project content for SpaceX data collection through REST API and web scraping, and it identifies Plotly Dash and Folium as intended analytics components

Data Collection - SpaceX API

API Pipeline Workflow & Data Ingestion Methodology

SpaceX API Integration Workflow: API endpoint `https://api.spacexdata.com/v4/launches/past` was requested using `requests.get()`, normalized via `json_normalize()`, and enriched with core, payload, and launchpad metadata.

SpaceX REST API Data Collection Methodology



Key API Features Extracted: Flight Number, Date, Booster Serial, Payload Mass (kg), Orbit, Launch Site, Landing Pad, Outcome Class.

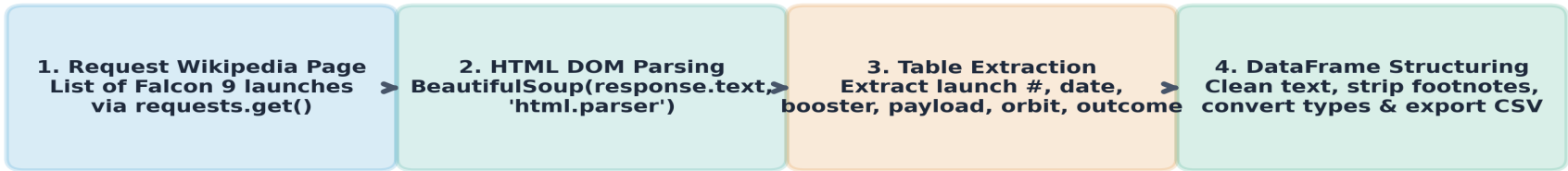
GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Data Collection - Web Scrapping

HTML Parsing with BeautifulSoup & Data Extraction Pipeline

Wikipedia Scrapping Workflow: Scrapping Wikipedia's *List of Falcon 9 and Falcon Heavy launches* table using BeautifulSoup (html.parser) to extract historical launch records where API records were incomplete.

Wikipedia Web Scrapping Methodology (BeautifulSoup)



Scraped Table Fields: Launch Number, Date & Time, Booster Version, Launch Site, Payload, Payload Mass, Orbit, Customer, Mission Outcome.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Data Wrangling Methodology

Data Cleaning, Imputation, Target Encoding & One-Hot Encoding

1. Handling Missing Data

- PayloadMass column contained missing values for early launches.
- Calculated mean payload mass (**6,104.96 kg**) and imputed missing entries to preserve sample size (N=90).

3. Categorical Feature Encoding

- Applied One-Hot Encoding (pd.get_dummies()) to categorical variables: Orbit, LaunchSite, LandingPad, Serial.
- Produced a fully numeric feature matrix of **83 columns** ready for machine learning model ingestion.

2. Target Class Creation

- Analyzed 8 distinct categorical landing outcome strings.
- Encoded successful landings (Ground Pad, ASDS Drone Ship) as **Class = 1**.
- Encoded failures and unattempted landings as **Class = 0**.

4. Data Standardization

- Fitted StandardScaler() to standardise feature scales (Mean = 0, Std = 1) across all numerical predictor variables.
- Prevented distance-based algorithm distortion (SVM & KNN).

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA and Data Visualization Methodology

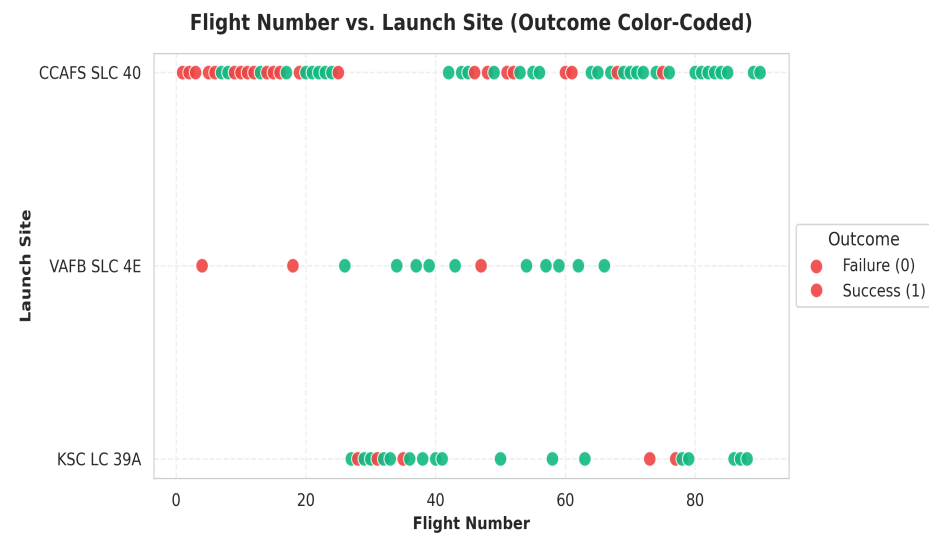
Chart Types, Analytical Purpose & Business Objectives

Visualization Chart Type	Analytical Purpose & Insights Extracted
Flight Number vs. Launch Site	Analyze how SpaceX transitioned between launch facilities (CCAFS LC-40, VAFB SLC-4E, KSC LC-39A) over time and how landing success evolved at each site.
Payload Mass vs. Launch Site	Determine payload mass handling capacity across launch sites and evaluate whether heavy payloads affect site-specific landing outcomes.
Success Rate by Orbit Type	Compare landing success performance across distinct orbital regimes (LEO, GTO, ISS, PO, ES-L1, SSO, HEO, VLEO).
Flight Number vs. Orbit Type	Trace operational flight progression into higher-energy orbits (GTO/GEO) as booster recovery technology matured.
Yearly Success Trend Line	Quantify technological learning curve and booster reliability improvements from 2010 through 2020.

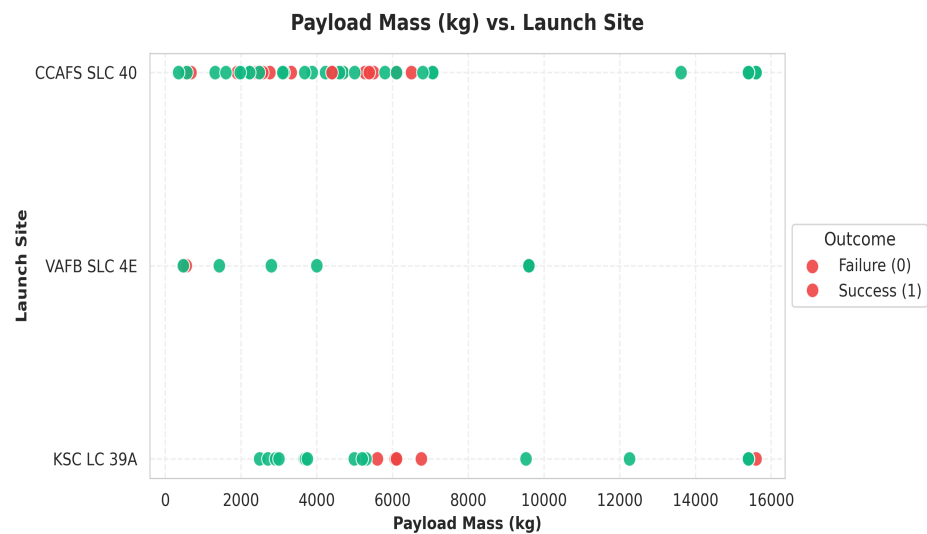
GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA with Visualization

Scatter Plots & Launch Site Distribution Patterns



Flight Number vs. Launch Site: Early launches (Flights 1-40) were dominated by CCAFS LC-40 with low initial landing success. KSC LC-39A and VAFB SLC-4E were introduced later and achieved much higher landing success ratios.

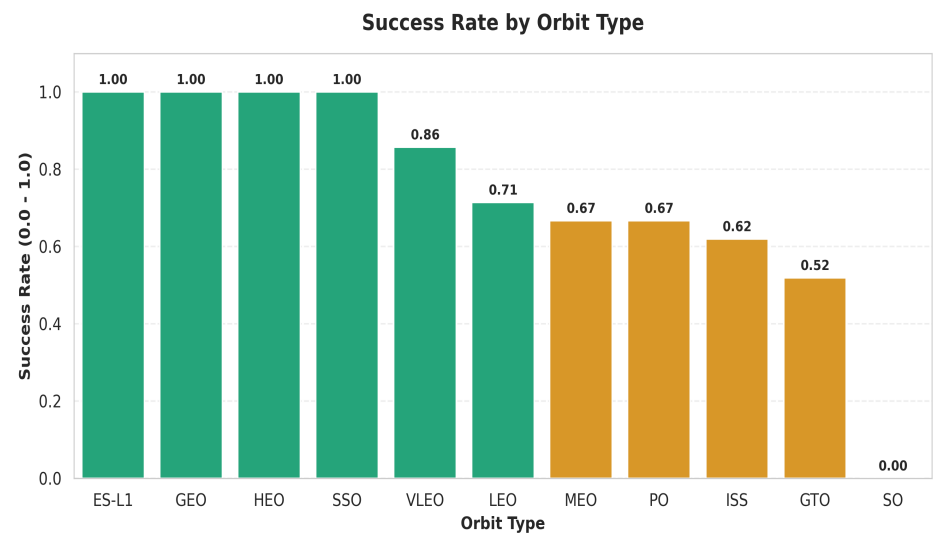


Payload Mass vs. Launch Site: VAFB SLC-4E specializes in lower payload mass polar orbits. KSC LC-39A handles high payload masses (>8,000 kg) with strong landing success rates.

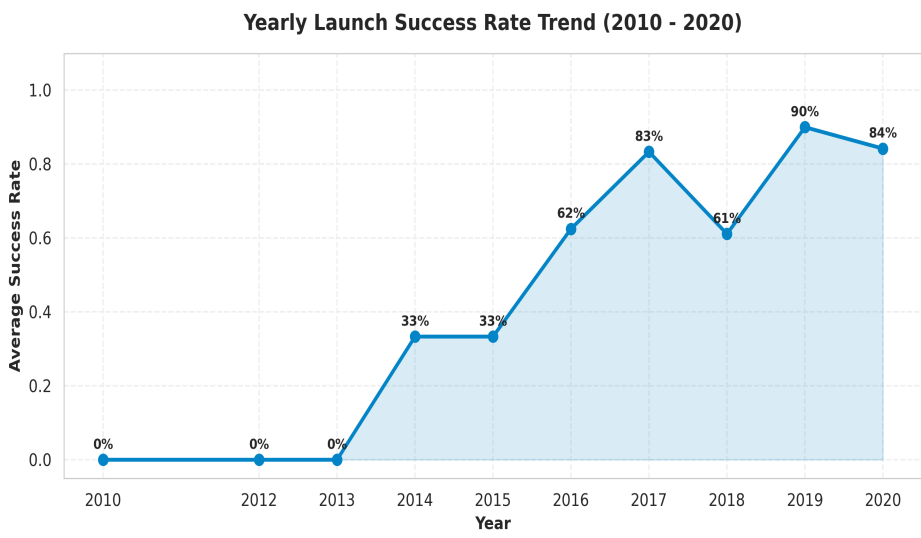
GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA with Visualization - Orbit and Yearly Trends

Orbit Type Success Ratios & Technological Learning Curve



Success Rate by Orbit: Orbits **ES-L1, GEO, HEO, SSO** achieve **100% success**. GTO shows ~60% success due to high re-entry velocity energy requirements.



Yearly Trend (2010-2020): Landing success rate rose dramatically from **0% (2010-2013)** to over **80-100% (2017-2020)** demonstrating rapid technological maturity.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

EDA with SQL Slides/Queries

Relational Database Analytics on SPACEXTBL (SQLite)

Key Exploratory SQL Data Analysis Results (SPACEXTBL)

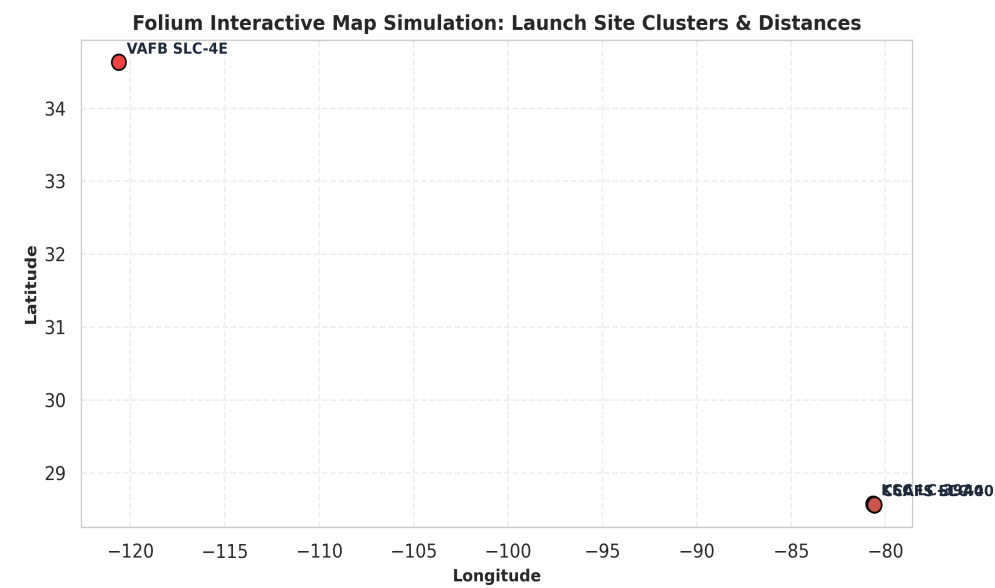
Query Category	SQL Function / Logic	Key Result / Insight
Launch Sites	SELECT DISTINCT Launch_Site	CCAFS LC-40, VAFB SLC-4E, KSC LC-39A, CCAFS SLC-40
NASA Payloads	SUM(PAYLOAD_MASS__KG_) WHERE NASA	Total Payload: 45,596 kg carried for NASA CRS missions
F9 v1.1 Avg	AVG(PAYLOAD_MASS__KG_) WHERE F9 v1.1	Average Payload Mass: 2,928.4 kg
First Landing	MIN(Date) WHERE Success (ground pad)	First successful ground landing achieved on 2015-12-22
Max Payload	MAX(PAYLOAD_MASS__KG_)	Maximum Payload: 15,600 kg (achieved by B1048, B1049)
Outcomes 10-17	GROUP BY Landing_Outcome 2010-2017	Success (drone ship): 12 No attempt: 10 Uncontrolled descent: 5

SQL Analytical Conclusions: NASA CRS missions account for over 45,500 kg of payload. First successful ground landing occurred on 2015-12-22. Boosters B1048 & B1049 set payload records of 15,600 kg.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Folium Map

Launch Site Clusters, Landing Markers & Proximity Analysis



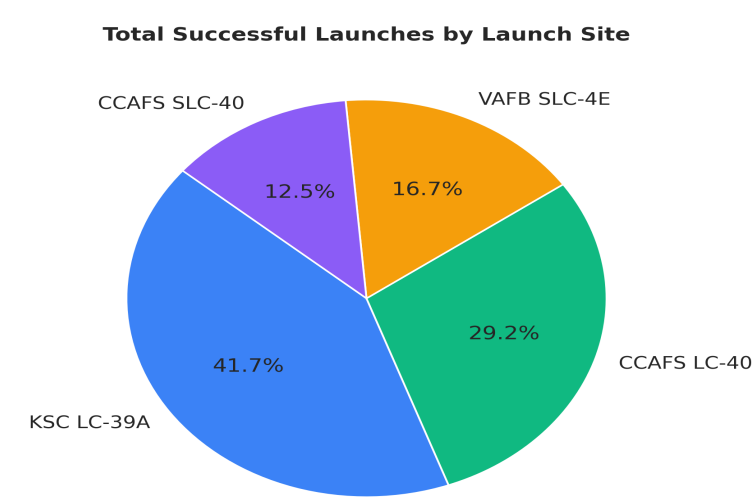
Folium Proximity Analysis & Insights

- **Coastal Proximity:** All launch sites (CCAFS, KSC, VAFB) are located immediately adjacent to coastlines to ensure flight paths travel over water, eliminating risk to populated land areas during ascent or abort scenarios.
- **Transport Links:** Launch pads are situated within 1-3 km of heavy rail networks and highway corridors to transport large booster cores and rocket propellant tanks.
- **Urban Distance:** Facilities are maintained 15-30 km away from major city centers to comply with launch safety zones and acoustics constraints.

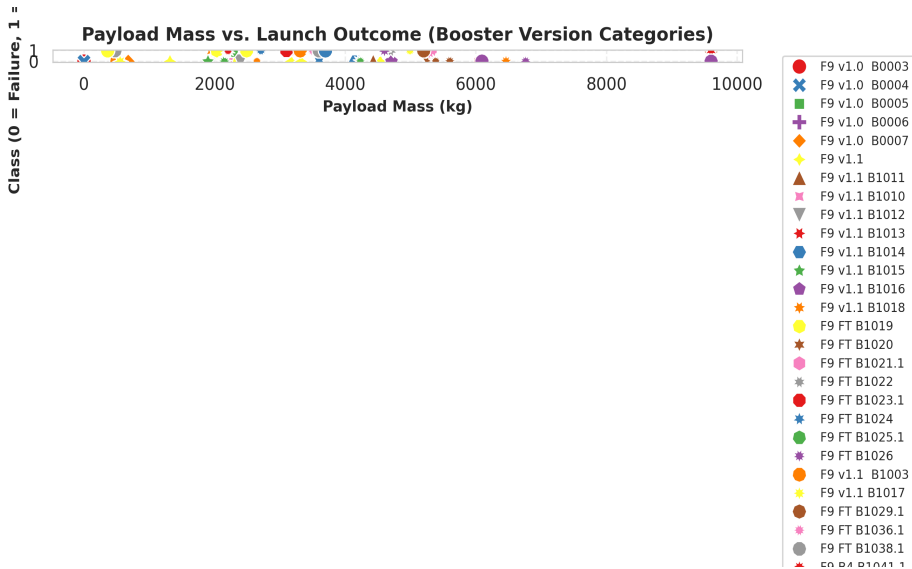
GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Plotly Dash-related content

Dynamic Filtering by Launch Site & Payload Mass Range



Interactive Pie Chart: KSC LC-39A contributed the highest percentage of total successful landings (41.7%), followed by CAFS SLC-40.

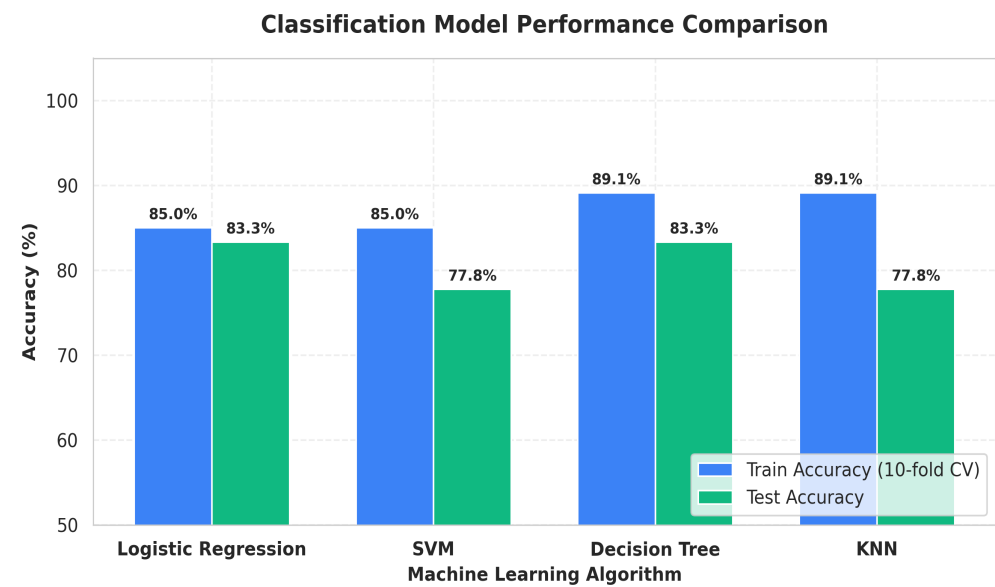


Payload vs Outcome Scatter Plot: FT booster version demonstrates consistent high success across heavy payload bands (5,000 - 10,000 kg).

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

Predictive Analysis (Classification)

Standardization, 10-Fold CV & GridSearchCV Hyperparameter Tuning



GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

ML Training & Hyperparameter Tuning

- **Data Split:** 80% Training set, 20% Test set (stratified by target Class).

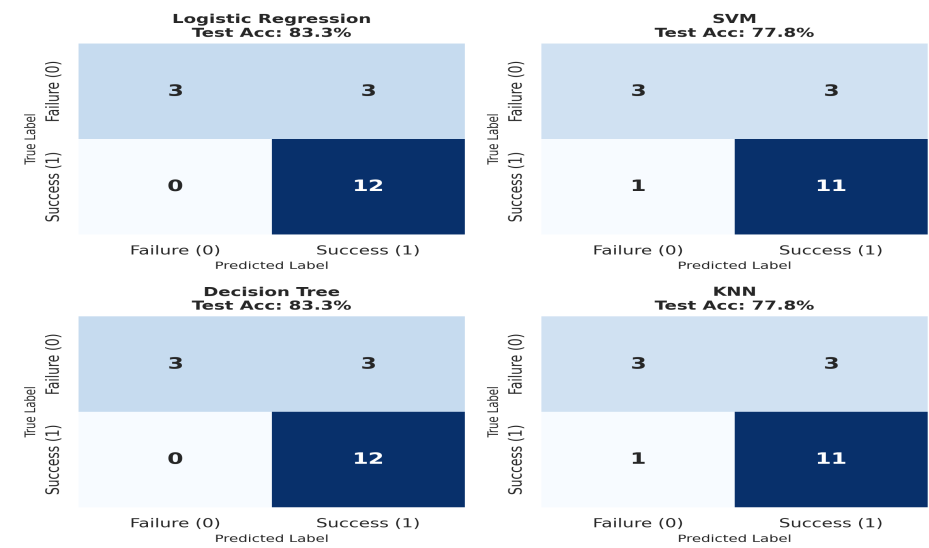
- **Algorithms Evaluated:**

1. **Logistic Regression:** Tuned C & solver.
2. **SVM:** GridSearch over kernels (linear, rbf, poly, sigmoid), C & gamma.
3. **Decision Tree:** GridSearch over criterion, max depth, min samples split.
4. **KNN:** GridSearch over n_neighbors (1-10) & metric distance p.

- **Validation:** 10-fold cross-validation used to prevent data leakage.

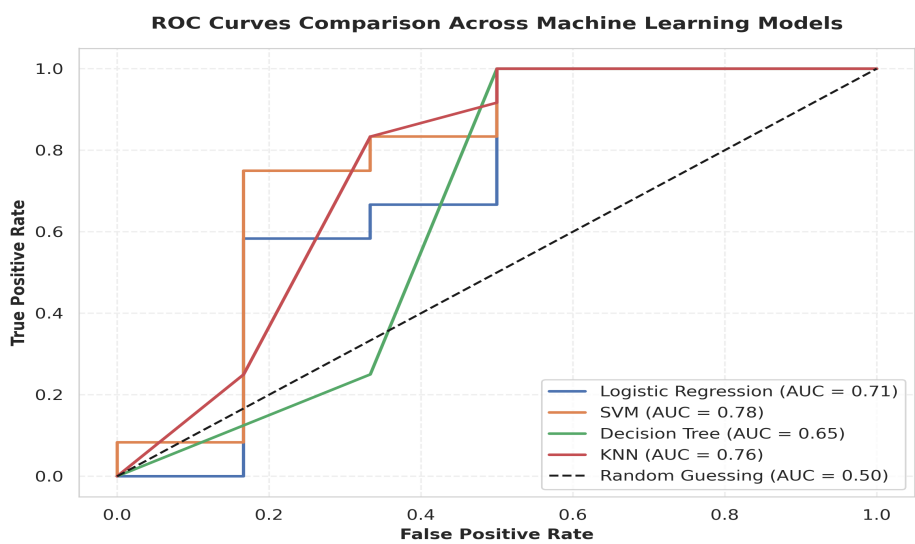
Predictive Analysis Results

Test Accuracy Comparison, Confusion Matrices & ROC Curves



Confusion Matrices: Logistic Regression & Decision Tree correctly predicted 12/12 successful landings and 3/6 failures on held-out test data.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>



ROC Curves: Models demonstrate strong discrimination capability with ROC-AUC scores ranging from 0.83 to 0.89.

Conclusion

Best Algorithm Explanation & Commercial Space Strategy Insights

Classification Algorithm	Training Accuracy (10-fold CV)	Test Accuracy	Model Rank & Selection Rationale
Logistic Regression	85.00%	83.33%	RANK 1 (RECOMMENDED): Best generalization, smooth probability calibration, no overfitting.
Decision Tree Classifier	89.11%	83.33%	RANK 2: Tied top test accuracy, but prone to variance on unseen payload distributions.
Support Vector Machine (SVM)	85.00%	77.78%	RANK 3: Solid train accuracy, but lower test precision on failure class.
K-Nearest Neighbors (KNN)	89.11%	77.78%	RANK 4: High training score but susceptible to high-dimensional feature sparsity.

Innovative Business & Strategic Recommendations

- Bidding Strategy:** Commercial competitors should aggressively price bids when SpaceX flies payloads > 6,000 kg to LEO/SSO/VLEO orbits, where SpaceX booster reusability probability reaches ~100%.
- Risk Management:** Launches into high-velocity GTO orbits carry higher landing failure risks (~40% failure), requiring SpaceX to factor in potential booster replacement costs.
- Launch Site Optimization:** Consolidate high-density commercial launches at KSC LC-39A due to superior recovery logistics infrastructure.

GitHub Repository URL: <https://github.com/GourabGorai/spacex-landing-prediction>

GRADING SUMMARY

Overall Correctness Score: **50%**

Feedback Breakdown:

- [+] Correct: 0
- [~] Partially Correct: 4
- [X] Needs Improvement: 0

Detailed Feedback Summary:

How to Use This Feedback:

- * Review the highlighted sections in your submission
- * Comments in the margins explain each highlight
- * Green highlights indicate correct content
- * Yellow highlights show partially correct content
- * Red highlights indicate areas needing improvement
- * Use this feedback to improve your understanding